



# MERIT EDUCATION GROUP

## What Is a Variable?

*I can identify what a variable is and explain how it represents an unknown quantity.*

Standard: 9.3.5.3

Standard: 9.3.5.7

Standard: 9.3.5.8

Day 1 · Unit 1 · Theme: Minnesota State Fair

### 1 BELL RINGER — WHITEBOARDS READY!

Write the expression for “a number increased by 6” and hold up your board.

Your response: \_\_\_\_\_

### 2 KEY VOCABULARY

**Variable** — A \_\_\_\_\_, usually a letter, represents an unknown or changing quantity.

\_\_\_\_\_

**Expression** — A math phrase made of numbers, variables, and operation \_\_\_\_\_ (no equals sign).

\_\_\_\_\_

**Coefficient** — The number \_\_\_\_\_ by a variable in a term (3 in  $3n$ ). \_\_\_\_\_

**Constant** — A term that is a number on its own, with \_\_\_\_\_ variable. \_\_\_\_\_

**Evaluate** — To find the \_\_\_\_\_ of an expression by replacing variables with numbers.

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### 3 DIRECT INSTRUCTION — TAKE NOTES (I DO)

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**Example:** A number  $n$  is tripled, then increased by 4. Write an algebraic expression.

• Step 1 — The unknown number is represented by the variable \_\_\_\_\_. ( $n$ )

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• Step 2 — “Tripled” means multiply by \_\_\_\_\_  $\rightarrow$  \_\_\_\_\_. ( $3, 3n$ )

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• Step 3 — “Increased by 4” means add \_\_\_\_\_  $\rightarrow$  \_\_\_\_\_. ( $4, 3n + 4$ )

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### 4 GUIDED PRACTICE — WORK TOGETHER (WE DO)

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•  $x$  decreased by 9. The product of 6 and a number  $y$ .

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### 5 CULTURE CONNECTION — MINNESOTA STORIES

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**Ojibwe** 🇺🇸 (Tribal Nations) — Ojibwe families harvest wild rice (manoomin) together from canoes, measuring and counting each harvest’s pounds of grain.

**Word problem:** An Ojibwe elder counts  $4x$  fish in one net and  $3x$  fish in another net. Write an expression for the total number of fish. If  $x = 7$  fish per net, how many fish are there total?

*Language fact: Ojibwemowin (Ojibwe): bezhig, niizh, niswi, niiwin (1-4)*

## 6 INDEPENDENT PRACTICE — YOU TRY!

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- Twice a number increased by 3 equals 15. Write an equation and find the number.
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## 7 EXIT TICKET — SHOW WHAT YOU KNOW

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Write an expression for “a number divided by 4, then increased by 7.”

Answer: \_\_\_\_\_

*Show your work for full credit.*